Reg. No.:

Name :

VIT

Vellore Institute of Technology

FINAL ASSESSMENT TEST LAB – November 2024

Programme	:	B.Tech. (Set-~~B~~) A	**Semester**	:	Fall 2024-25
Course Title	:	Probability and statistics	**Code**	:	BMAT202P
			Slot	:	L37+L38
Class No.	:	CH2024250102409	**Max. Marks**	:	50
Faculty (s)	:	Dr. Amit kumar rahul	**Time**	:	90 min

Answer ALL the Questions

Q. No.	Sub. Sec.	Question Description	Marks

1. From the following data

Weekly wages (in hundred rupees)	10-15	15-20	20-25	25-30	30-40	40-60	60-70
Number of workers	7	19	27	15	12	12	8

10

(a) Represent the above data by means of a histogram. [2 Marks]
(b) Calculate mean for number of workers. [2 Marks]
(c) Calculate the standard deviation for workers. [2 Marks]
(d) Calculate the variance for workers. [2 Marks]
(e) Calculate the range of weekly wages. [2 Marks]

2 After conducting an analysis, it was determined that the demand for smartphones in a region largely depends on the population of the region. The table below shows the population (in lakhs) and smartphone sales (in '1000 units) for five regions in 2021:

Region	Population (in lakhs): X	Smartphone Sales (in '1000 units): Y
Hyderabad	50	18.2
Chennai	60	22.6
Coimbatore	55	19.8
Madurai	40	14.5
Vizag	70	26.7

10

(a) Fit a linear regression equation of Y on X and X on Y. [5 Marks]
(b) Determine the coefficients of the linear regression model. [2 Marks]
(c) Predict the smartphone sales for the region Delhi in 2022 if the population is expected to grow to 75 lakhs, assuming the same trend continues. [3 Marks]

3 A sample of 400 items is taken from a population whose standard deviation is 10 .The mean of the sample is 40. Test whether the sample has come from a population with mean 38. Also calculate 95% confidence limits for the population. 10

4 **Debug the CODE below and write the corrected version by highlighting the same:**

Comparing two independent sample means, taken from two populations with unknown variance. The following data shows the height. So find individuals of two different countries with unknown population variances. Is there any significant difference between the average heights of two groups?

X	20	13	16	19	15	18	18	17	17	19
Y	18	19	13	17	18	16	17	14	15	18

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```
x1=c()
x2=c()
n1=length()
n2=length()
x1bar=mean(x1)
x2bar=mean(x2)
sd1=sqrt(var)
sd2=sqrt(var)
t=abs(bar-x2)/(((sd1^2/n1)+(sd2^2/n2)))
tv=qt(1-(alpha/2),n1+-2)
```

5 **Answer the followingmultiple choice questions (each question carries one mark)**

1. Which command used for storing data tables in R-studio.
   a. Data.frame[]
   b. data.frame()
   c. Dataframe[]
   d. Frame.data()
2. Syntax for creating two-way table in R-studio.
   a. table(a, b)
   b. table[a,b]
   c. Table(), table()
   d. Two-table[]
3. What is the command for selecting 'n' numbers at a random from the set 1 to N?
   a. sample(1:N, n)
   b. Sample[n]
   c. nsample(1:N)
   d. Nsample(n)

10

4. What is the command for invoking packages in R-studio?
   a. library(package name)
   b. Package name(library)
   c. Library[package name]
   d. Package library (name)
5. What is command for finding t-distribution table value in R-studio?
   a. qt(1-alpha, df=n-1)
   b. pt(1-alpha, df=n-1)
   c. qt[alpha, df=n-1]
   d. pt[alpha, df=n-1]